

Project Title	Marketing campaign analysis
Skills take away From This Project	Python: Data cleaning, EDA, feature engineering, and CSV handling. SQL: Table design, loading data, analytical queries, KPIs. EDA & Visualization: Univariate, bivariate analysis, segmentation, and insights using either Streamlit(mandatory) or Power BI(Optional).
Domain	Marketing Analytics / Customer Analytics

Problem Statement:

A retail company has run multiple marketing campaigns and recorded detailed information about customers, their demographics, spending across product categories, channel-wise purchases, and whether they accepted specific campaigns.

Management wants a consolidated analytics solution to understand who its best customers are, what drives campaign acceptance, and how different customer segments behave across products and channels.

Learners must:

- Clean and analyze the given marketing data.
- Build a SQL data model and write analytical queries.
- Create an interactive dashboard (Streamlit app or Power BI report) to answer key business questions.

Business Use Cases:

- Which customer segments have the highest response rate to campaigns (overall and per campaign: AcceptedCmp1–5, Response)?

- How do spending patterns across products (wine, fruits, meat, fish, sweets, gold) vary by age, income, marital status, and country?
- Which channels (web, store, catalog, deals) are most used by high-value customers, and how often do they visit the website?
- Are there specific segments that are under-served (low spending, high visits, low response)?
- What are characteristics of “ideal target customers” for future campaigns? (age range, income band, family composition, country, etc.)

Approach:

1. Data Understanding & Cleaning (Python)

- Load marketing_data.csv.
- Use marketing_data_dictionary.csv to understand fields and meanings.
- Handle missing values, data types (dates, integers, categorical), outliers, and derived columns (age, total spend, total purchases, total children, etc.).

2. Exploratory Data Analysis (Python)

- Perform univariate analysis: distributions of age, income, spending, recency, etc.
- Perform bivariate/multivariate analysis: response vs. income, response vs. age, response vs. product spend, channel usage vs. response, etc.
- Identify key patterns and segments.

3. Rule-Based Segmentation (Python & SQL)

- Define segmentation criteria using business rules (see table below).
- Implement segments in Python using if-else logic or pd.cut().
- Summarize segments in SQL using CASE statements.
- Visualize and interpret segment-wise KPIs in the dashboard.

4. SQL Data Modeling & Queries

- Create database schema (at minimum a main customer table; optionally derived tables for segments or campaign KPIs).
- Load cleaned data into SQL.
- Write queries to:
 - i. Compute KPIs (response rate, average spend per segment, average visits, etc.).
 - ii. Generate segment-level summaries (by education, marital status, country, age band, income band, etc.).
 - iii. Support dashboard requirements (fact tables / views).

5. Dashboard (Streamlit or Power BI)

- Connect to the SQL layer or clean CSV.
- Build visuals and filters for key business questions and KPIs.
- Allow users (stakeholders) to explore segments and campaign performance interactively.

Results:

Clearly describe who the most valuable and most responsive customer segments are, with evidence from metrics and charts.

Present a working dashboard where stakeholders can explore customer profiles, spending patterns, and campaign responses using filters and visuals.

Provide 4–6 actionable recommendations for improving future campaigns (e.g., which segments to target, which channels to prioritize).

Project Evaluation metrics:

Area	Example criteria
Python & EDA	Data cleaning completeness, correct derived features, depth of EDA, code quality.
SQL	Schema design, correctness and complexity of queries, use of aggregations/window functions.
Dashboard	Relevance of visuals, interactivity, layout and usability, clarity of insights.

Business Insight	Quality of interpretations, recommendations linked to data, storytelling.
Documentation	Project report clarity, explanation of choices, reproducibility.

Technical Tags:

Python, Pandas, NumPy, SQL, EDA, Data Cleaning, Data Visualization, Customer Segmentation, Marketing Analytics, Streamlit, Power BI, Dashboard, KPI Analysis.

Data Set:

[marketing_data.csv](#): Main dataset with customer-level information.

[marketing_data_dictionary.csv](#): Field names and descriptions.

Key variables (non-exhaustive):

- Demographics: ID, Year_Birth, Education, Marital_Status, Income, Kidhome, Teenhome, Country.
- Customer relationship: Dt_Customer, Recency.
- Spending: MntWines, MntFruits, MntMeatProducts, MntFishProducts, MntSweetProducts, MntGoldProds.
- Channel usage: NumDealsPurchases, NumWebPurchases, NumCatalogPurchases, NumStorePurchases, NumWebVisitsMonth.
- Campaigns: AcceptedCmp1–AcceptedCmp5, Response.
- Other: Complain.

Data Set Explanation:

- Each row represents a unique customer, identified by ID.
- The dictionary file defines each field and its meaning, including campaign acceptance flags and spending measures over the last two years.

Minimum preprocessing expectations:

- Convert Dt_Customer to date type and derive Customer_Tenure (e.g., in days or months from reference date).

- Create derived metrics:
 - Age from Year_Birth.
 - Total_Spend as sum of all Mnt* fields.
 - Total_Purchases as sum of channel purchases.
 - Children = Kidhome + Teenhome.
- Handle outliers and missing values in numeric fields (e.g., unrealistic ages, income).

Rule based segmentation task:

Segment Name	Rule Definition	Example Value(s)
High Income	Income > ₹75,000	75001–99999
Young Customer	Age < 30	18–29
Campaign Responder	Response = 1	1
High Web Engagement	NumWebVisitsMonth > 5	6–20
Family Customer	Children > 0	1–3
High Spender	Total_Spend > 90th percentile	> ₹50,000

Project Deliverables:

1. Python Notebook(s)
 - Data loading, cleaning, EDA, feature engineering, and segmentation.
 - Clear commentary and visualizations.
2. SQL Scripts
 - DDL to create tables.
 - DML to insert/load data.
 - Analytical queries and views used for the dashboard.
3. Dashboard (choose one):
 - Streamlit app: project structure, `app.py`, and any config files.
 - Power BI: `.pbix` file with all visuals and measures.
4. Project Report (PDF / Doc)

- Problem understanding, approach, key findings, and recommendations.
5. Readme
- Setup instructions and how to run the notebook and dashboard.

Project Guidelines:

- Coding standards:
 - Use functions where appropriate and meaningful variable names.
 - Organize code into logical sections (loading, cleaning, EDA, modeling/derivations, export).
- Version control:
 - Prefer Git for version history where possible.
- SQL best practices:
 - Use primary keys, proper data types, and normalized design where feasible.
 - Use comments in SQL scripts to explain non-trivial queries.
- Dashboard best practices:
 - Use clear titles, labels, and legends; avoid clutter.
 - Include filters (e.g., Country, Education, Marital_Status, Age band, Income band).
- Interpretation:
 - Every major chart or table should have a brief takeaway in the report or dashboard tooltips.

Reference:

*****If you don't know how to approach the project, kindly refer to the project orientation recording provided in this table. (Available in English & Hindi)***

Streamlit Doc	https://docs.streamlit.io/library/api-reference
SAMPLE PROJECT FOR REFERENCE(ENGLISH)	 Project Excellence Series: Guided...

Streamlit recording (English)	Special session for STREAMLIT(11/08/2...
EDA Guide	Exploratory Data Analysis (EDA) Guide
Project Live Evaluation Metrics	Project Live Evaluation
Capstone Explanation Guideline	Capstone Explanation Guideline
GitHub Reference	GitHub Project Submission Guide
PowerBi Session Recording Link	PowerBi by Lokesh
PowerBi and SQL connection	Connect MySQL to PowerBI
Steps to Create a Free Power BI Work Account	Steps to Create a Free Power BI Work A...
POWERBI_MYSQL_CONN_Issue	POWERBI_MYSQL_CONN_ISSUE
Project Orientation (English)	Project Orientation Session : Marketing ...
Project Orientation (Hindi)	Project Orientation Session : Marketing ...

Timeline:

The project must be completed and submitted within **10 days from the assigned date.**

PROJECT DOUBT CLARIFICATION SESSION (PROJECT AND CLASS DOUBTS)

About Session: The Project Doubt Clarification Session is a helpful resource for resolving questions and concerns about projects and class topics. It provides support in understanding project requirements, addressing code issues, and clarifying class concepts. The session aims to enhance comprehension and provide guidance to overcome challenges effectively.

Note: Book the slot at least before 12:00 Pm on the same day

Timing: Monday to Saturday (3:30 PM to 4:30 PM)

Session link and guidelines: [Project Doubt Session \[DS/AIML\]](#)

LIVE EVALUATION SESSION (CAPSTONE AND FINAL PROJECT)

About Session: The Live Evaluation Session for Capstone and Final Projects allows participants to showcase their projects and receive real-time feedback for improvement. It assesses project quality and provides an opportunity for discussion and evaluation.

Note: This form will Open on Saturday and Sunday Only on Every Week

Timing: Monday-Saturday (5:30 PM to 6:30 PM)

Booking link : <https://forms.gle/1m2Gsro41fLtZurRA>

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